Inventory Stock Management System

Using Data Structures and Algorithms

Project Report and Synopsis

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1. Introduction

Inventory tracking maintains accurate quantities and records of goods. This project offers a small browser-based system for product records and stock movement.

2. Problem statement

Manual tracking can create duplicate records, stock-count errors, and delayed replenishment.

3. Objectives

Maintain product catalog; record inward/outward transactions; prevent invalid stock changes; report low stock and valuation; demonstrate DSA.

4. Architecture

Browser UI → Flask routes → SQLAlchemy ORM → SQLite database. A separate C++17 file demonstrates DSA.

5. Functional requirements

Login; product search/add/delete/sort; stock inward/outward; low-stock list; reports; category and supplier pages.

6. Database schema

User(id, username, password_hash); Category(id, name); Supplier(id, name, company_name, contact_person, phone, email, address); Product(id, name, sku, quantity, purchase_price, selling_price, min_stock_level, unit, description, date_added, category_id, supplier_id); StockTransaction(id, product_id, transaction_type, quantity, note, created_at).

7. DSA concepts

Linear SKU search O(n); binary search on sorted names O(log n); sorting typically O(n log n); C++ unordered_map average O(1) lookup; queue FIFO; stack LIFO.

8. Testing

Unit tests cover SKU search, binary search, sorting, queue and stack. Manual tests: duplicate SKU rejected, outward stock cannot exceed quantity, invalid quantity rejected, low-stock alert appears at threshold.

9. Limitations

Basic demo login, reference-only categories/suppliers, no roles, and DSA undo stack does not reverse committed DB transactions.

10. Future scope

Role access, CSV/Excel export, purchase orders, barcode scanning, and transaction reversal with audit trail.

11. Conclusion

The application demonstrates inventory workflows, data modeling, validation, reports, and fundamental DSA concepts.

Project Synopsis

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Stack: Python, Flask, SQLite, HTML, CSS, JavaScript, C++17.

Abstract

This web application maintains product details, tracks stock inward/outward, reports low-stock items, and stores transaction history in SQLite. DSA demonstrations include search, sorting, a FIFO queue, a LIFO stack, and hash-based SKU lookup.

Objectives

• Centralize product records and stock levels.

• Validate stock movement.

• Flag low-stock items.

• Demonstrate searching, sorting, queues, stacks, and hashing.

Scope

Local academic demo, seeded user/data, not production deployment.